Actsc 372 Investment Science and Corporate Finance

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0 The Role of Financial Intermediaries

Lecture 1, 2025/09/03

What is financial investment?

- Ownership / loan contracts helped by financial intermediaries.

Role of Financial Intermediaries

- **Size intermediation**: the bank gets small lenders together with the “big” borrowers.
- **Term intermediation**: between short term lenders and long term borrowers.
- **Risk intermediation**: pooling resources reduces individual risk.

Consumer Smoothing: see handwritten notes.

1 Investment Rules

1.1 The NPV Rule

Question: Given a choice of projects, what project should a rational investor choose?

Definition 1.1 (**Project**).

A **project** is a collection of cash flows $\{C_0, C_1, C_2, \ldots\}$ where C_t = cash flow at time t.

Note.

- $C_t < 0 \implies$ cost.
- $C_t > 0 \implies$ revenue.

Definition 1.2 (**Pure Investment Project**).

A **pure investment project** is a project where $C_0 < 0$ and $C_t \geq 0$ for all $t > 0$.

Definition 1.3 (**Pure Financing Project**).

A **pure financing project** is a project where $C_0 > 0$ and $C_t \leq 0$ for all $t > 0$.

Definition 1.4 (**Independent Projects**).

A and B are **independent projects** if the cash flows of A do not affect the cash flows of B.

Note. Between projects A and B, we can choose:

- one,
- both,
- or neither.

Definition 1.5 (**Mutually Exclusive Projects**).

A and B are **mutually exclusive projects** if choosing $A \implies$ we cannot choose B and vice versa (we cannot choose both).

Example. Given $C_0, C_1, C_2, \ldots$. How to decide if this is a "good" project?

Proof. See NPV rule below. $\square$

Definition 1.6 (**Net Present Value (NPV)**).

The **NPV** of a project is

$$NPV = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots = \sum_{t=0}^{\infty} \frac{C_t}{(1+r)^t}$$

where units = dollars.

Definition 1.7 (**NPV Rule**).

- For independent projects, choose all projects with $NPV > 0$.
- For mutually exclusive projects, choose the project with the highest NPV, as long as $NPV > 0$.

Example. Let $C_0 = -100, C_1 = 110, C_t = 0$ for $t \geq 2$, and $r = 8\%$. Then,

$$NPV = -100 + \frac{110}{1.08} > 0.$$

So, we should accept this project.

Example. Let $C_0 = 100, C_1 = -110$, and $r = 8\%$. Then,

$$NPV = 100 - \frac{110}{1.08} < 0.$$

So, we should reject this "bad" project.

What is r? $\implies$ r = "opportunity cost" of the dollar.

Lecture 2, 2025/09/08

Advantages of NPV rule:

- NPV takes into account the time value of money.

- NPV takes into account cash flows, not accounting incomes.

Example. Let $C_0 = 100, C_1 = 115, r = 10\%$. The project has positive NPV and the owner will be happy if the project is taken.

Disadvantages of NPV rule:

- The cash flows are uncertain.
- What should be the right "r"? (see later: CAPM)

1.2 Review of ACTSC 231

Some important cash flows to recall from ACTSC 231:

- Propetuity.
- Propetuity with growth.
- Annuity.

Definition 1.8 (Propetuity, Perpetuity Factor).
A **propetuity** is an annuity with no end dates (infinite # of payments). Let $C_t = C$ for all $t \geq 1$. Then,

$$\begin{aligned}
PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \cdots \\
&= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots\right] \\
&= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1}{1+r}} \\
&= \frac{C}{r}.
\end{aligned}$$

The **perpetuity factor** is $\frac{1}{r}$.

Definition 1.9 (Perpetuity with Growth).

Let $C_t = C(1+g)^{t-1}$ for $t \geq 1$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C(1+g)}{(1+r)^2} + \frac{C(1+g)^2}{(1+r)^3} + \cdots \\ &= \frac{C}{1+r}\left[1 + \frac{1+g}{1+r} + \left(\frac{1+g}{1+r}\right)^2 + \cdots\right] \\ &= \frac{C}{1+r} \cdot \frac{1}{1 - \frac{1+g}{1+r}} \\ &= \frac{C}{r-g}, \quad g < r. \end{aligned}$$

Definition 1.10 (Annuity, Annuity Factor).

Let $C_t = C$ for $1 \leq t \leq n$. Then,

$$\begin{aligned} PV &= \frac{C}{1+r} + \frac{C}{(1+r)^2} + \cdots + \frac{C}{(1+r)^n} \\ &= \frac{C}{1+r}\left[1 + \frac{1}{1+r} + \cdots + \frac{1}{(1+r)^{n-1}}\right] \\ &= \frac{C}{1+r} \cdot \frac{1 - \left(\frac{1}{1+r}\right)^n}{1 - \frac{1}{1+r}} \\ &= C \cdot \frac{1-(1+r)^{-n}}{r}. \end{aligned}$$

The **annuity factor** is $\frac{1}{r}\left[1 - (\frac{1}{1+r})^n\right]$.

1.3 Other Investment Rules

Definition 1.11 (Payback Period).

The **payback period** is the # of years required to recoup the initial investment (ignoring the time value of money).

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$. The payback period is 3 years.

Note. The units of payback period is years.

Definition 1.12 (Payback Period Rule).

Decide on a cutoff period, T^*. If the payback period $\leq T^*$, accept; otherwise, reject it.

Disadvantages of Payback Period Rule:

- It does not take into account the time value of money.
- The cutoff is arbitrary.
- Negative NPV may be accepted.
- Positive NPV may be rejected.

Definition 1.13 (Discounted Payback Period).

The **discounted payback period** is the payback period, but without ignoring the time value of money.

Example. Let $C_0 = -300, C_t = 100$ for $t \geq 1$, and $r = 12.5\%$. We are solving $\sum_{t=1}^{T} \frac{C_t}{(1+r)^t} = -C_0$. This implies that the discounted payback period, T, is 4 years.

Definition 1.14 (Internal Rate of Return (IRR)).

The **internal rate of return** (**IRR**) of a project is the cost of capital such that the NPV is zero.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. We have two ways to determine if we should accept this project:

(1) Calculate NPV: $NPV = -100 + \frac{112}{1.1} > 0 \implies$ accept.

(2) The project has a return of 12%. Since it is bigger than $r = 10\%$, we should accept it.

Question: How to solve for the IRR?

Answer: Calculate the NPV and equal it to zero, then solve for r^*. That is,

$$\sum_{t=0}^{T} \frac{C_t}{(1+r^*)^t} = 0.$$

Definition 1.15 (**IRR Rule**).

Calculate the iRR of a project.

- For independent projects, choose all projects with IRR $> r$.
- For mutually exclusive projects, choose the project with the highest IRR (as long as IRR $> r$).

Advantages of IRR rule:

- In order to calculate IRR, we do not need r.
- It usually agrees with the NPV rule.

Disadvantages of IRR rule:

The first two disadvantages are for independent projects.

(1) IRR cannot distinguish between investment and financing.

Note. $(-1)\sum \frac{C_t}{(1+r^*)^t} = 0(-1)$, the solution remains the same.

Example. Let $C_0 = -100, C_1 = 112, r = 10\%$. The IRR is 12% and we should accept this project.

Let $C_0 = 100, C_1 = -112, r = 10\%$. The IRR is still 12% but we should reject this project.

(2) Polynomial equation problem: finding the IRR = solving a polynomial equation, i.e.

$$C_0 + \frac{C_1}{1+r^*} + \frac{C_2}{(1+r^*)^2} + \cdots + \frac{C_T}{(1+r^*)^T} = 0.$$

However, we could have no roots, multiple roots, imaginary roots, or complex roots.

Example. Let $C_0 = 0, C_1 = 100, C_t = 0$ for all $t \geq 2$. The IRR $= \infty$.

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$. Then,

$$-100 + \frac{230}{1+r^*} - \frac{132}{(1+r^*)^2} = 0 \implies r^* = 10\% \text{ or } 20\%.$$

This is problematic, since which IRR should we use?

We will need the following result to deal with the multiple IRR problem.

Theorem 1.1 (**Descartes' Rule of Signs**).

The # of positive roots of a polynomial is $\leq$ # of "sign switches" of the coefficients.

Example. For the previous example where $C_0 = -100, C_1 = 230, C_2 = -132$, there are 2 sign switches. Thus, there are at most 2 positive roots.

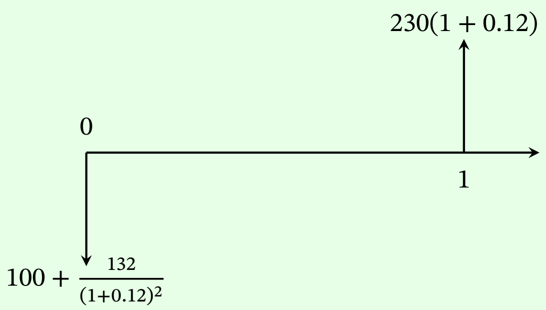
We have the following MIRR to avoid multiple solutions.

Definition 1.16 (**Modified Internal Rate of Return (MIRR)**).

The **modified IRR**

- brings all negative cash flows to time $t = 0$, and
- brings all positive cash flows to time $t = T$,

Example. Let $C_0 = -100, C_1 = 230, C_2 = -132$, and $r = 12\%$. Then,



Calculate the IRR of the modified project (MIRR):

$$-\left(100 + \frac{132}{(1+0.12)^2}\right) + \frac{230(1+0.12)}{1+r^*} = 0 \implies r^* = 12.36\%.$$

Note. We only have 1 "sign switch" in the modified project, so MIRR is unique. Then, we can compare MIRR with r, and choose the project with MIRR $> r$. But we need to know today's interest rate r.

Disadvantages of IRR rule (continued):

For mutually exclusive projects (i.e. we can only choose one project.):

(3) Scaling problem: the IRR is biased towards "smaller" projects.

Example.

Project A: $C_0 = -1, C_1 = 2$.

Project B: $C_0 = -100, C_1 = 150$.

Note that $\text{IRR}_A = 100\%$ and $\text{IRR}_B = 50\%$. However, project B is a better one.

Note. $\sum \frac{kC_t}{(1+r)^t} = 0$. IRR remains the same if you multiply by a constant.

Lecture 3, 2025/09/10

Here are some ways to deal with the scaling problem:

- Calculate the NPV and choose the higher one. In the example, $\text{IRR}_A > \text{IRR}_B$, but $\text{NPV}_A = 1 < \text{NPV}_B$.
- If the IRR picks the "smaller" project, then we do **incremental analysis**.

Example. Incremental project: $C = B - A$ with $C_0 = -99, C_1 = 148$.

If the IRR of the incremental project $< r \implies$ stick with the smaller project. If $> r \implies$ go with the bigger project.

(4) Timing problem.

Consider pure investment projects.

Example.

Project short: $C_0 = -300, C_1 = 345, C_t = 0$ for $t \geq 2$.

Project long: $C_0 = -200, C_4 = 330, C_t = 0$ for $t \neq 0, 4$.

Note that

$$IRR_{\text{short}} = 15\%, \quad NPV_{\text{short}} = 13.64$$

$$IRR_{\text{long}} = 13.34\%, \quad NPV_{\text{long}} = 25.34.$$

Example.

IRR assumes that all cash flows can be reinvested at the IRR itself.

A more realistic assumption that cash flows can be reinvested at r, not IRR.

Definition 1.17 (Profitability Index (PI)).

The **profitability index** is

$$PI = \frac{PV \text{ of future cash flows}}{\text{Initial Investment}} = \frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0}.$$

Definition 1.18 (PI Rule).

- For independent projects, choose all projects with $PI > 1$.
- For mutually exclusive projects, choose the project with the highest PI with $PI > 1$.

Proposition 1.2. For independent projects, PI agrees with the NPV rule.

Proof. Note that

$$\frac{\sum_{t=1}^{\infty} \frac{C_t}{(1+r)^t}}{-C_0} > 1 \implies NPV > 0.$$

□

Disadvantages of PI rule:

For mutually exclusive projects:

(1) Scaling problem (same as IRR).

Example. Let $r = 10\%$.

Project A: $C_0 = -10, C_1 = 15, C_2 = 40$.

Project B: $C_0 = -20, C_1 = 70, C_2 = 10$.

Note that

$$PI_A = 4.53, \quad NPV_A = 35.3$$
$$PI_B = 8.52, \quad NPV_B = 50.5.$$

Summary of Investment Rules: NPV rule is the best.